Urban Heat Island Analysis of Ann Arbor, MI
A Data-Driven Look at Heat, Tree Canopy, and Planting Priorities in Ann Arbor

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Introduction

- Urban Heat Islands (UHIs) occur when roads, buildings, and other impervious surfaces absorb and re-radiate more heat than natural landscapes.
- UHIs contribute to higher temperatures, increased energy demand, heat-related health risks, and impacts on water quality (EPA).
- Land Surface Temperature (LST), measured using satellite thermal imagery, provides a way to identify and map urban heat patterns.
- This project analyzed the spatial distribution of LST across Ann Arbor, Michigan using ArcGIS Pro and ArcGIS Online.
- Landsat 8 & 9 thermal imagery, NLCD tree canopy and impervious surface data, SimplyAnalytics demographic data, and City of Ann Arbor GIS datasets were integrated.
- Three interactive dashboards (census block, block group, and neighborhood scales) visualize environmental, demographic, and municipal factors related to urban heat.
- The dashboards allow users to compare tree canopy, impervious surfaces, population characteristics, parks, street trees, and publicly owned land in relation to heat patterns.
- This work demonstrates how GIS can support heat mitigation planning, urban forestry, sustainability initiatives, and climate resilience.

Area of Study

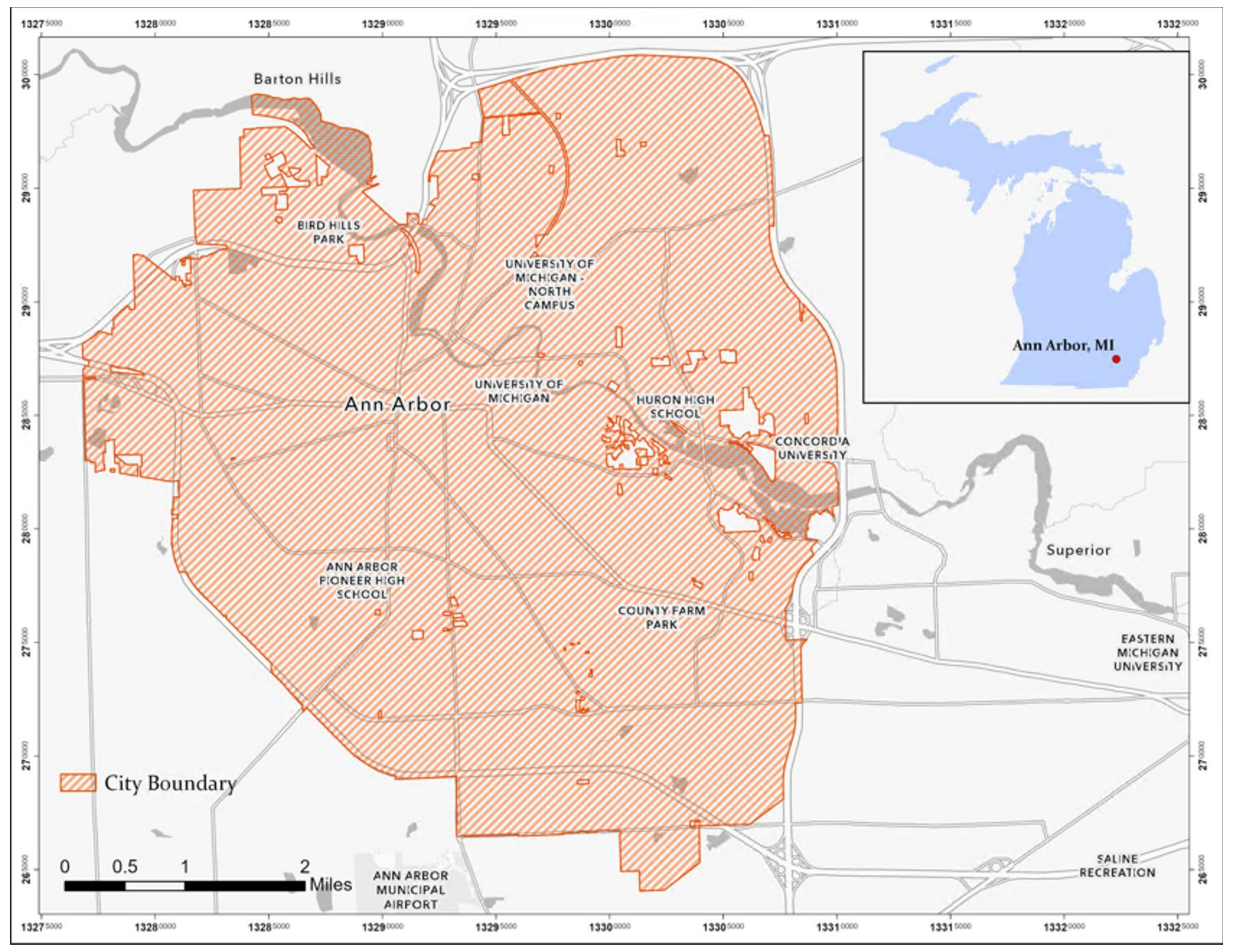

Figure 1. City of Ann Arbor municipal boundary and study area

Methods

- Collected Landsat 8 & 9, NLCD, SimplyAnalytics, U.S. Census, and City of Ann Arbor GIS datasets.
- Preprocessed raster data in Google Earth Engine and completed spatial analysis in ArcGIS Pro.
- Calculated average Land Surface Temperature (LST), tree canopy cover, and impervious surface coverage using Zonal Statistics.
- Integrated demographic and municipal datasets through spatial joins to examine environmental and community characteristics associated with urban heat.
- Developed a Priority Planting Score by combining LST, tree canopy, impervious surface, and street tree density to identify potential tree planting locations.
- Published three interactive ArcGIS Online dashboards at the census block, block group, and neighborhood scales to visualize results.

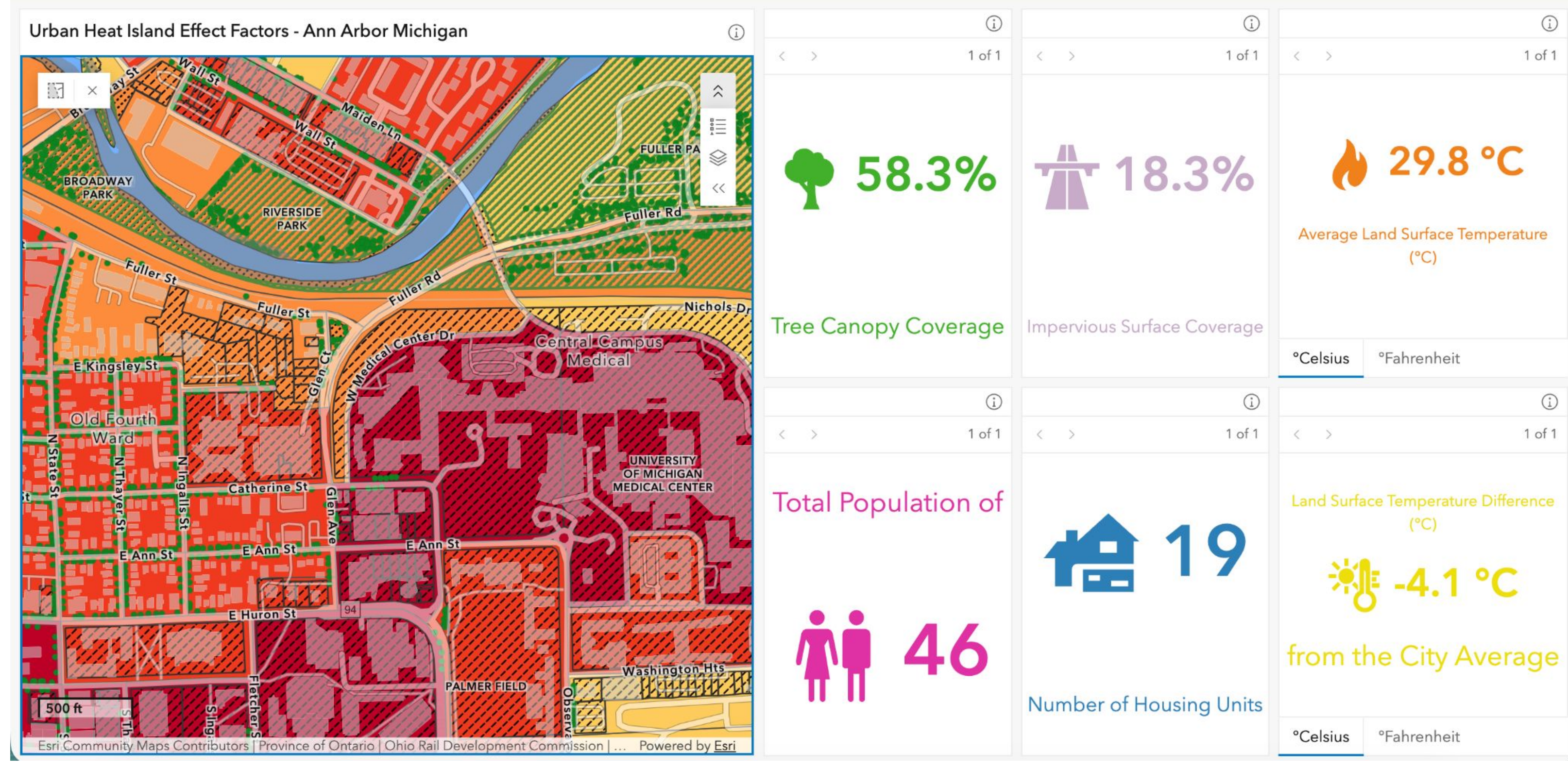

Fig. 2 Census block level interactive dashboard displaying land surface temperatures, population and housing

Results & Analysis

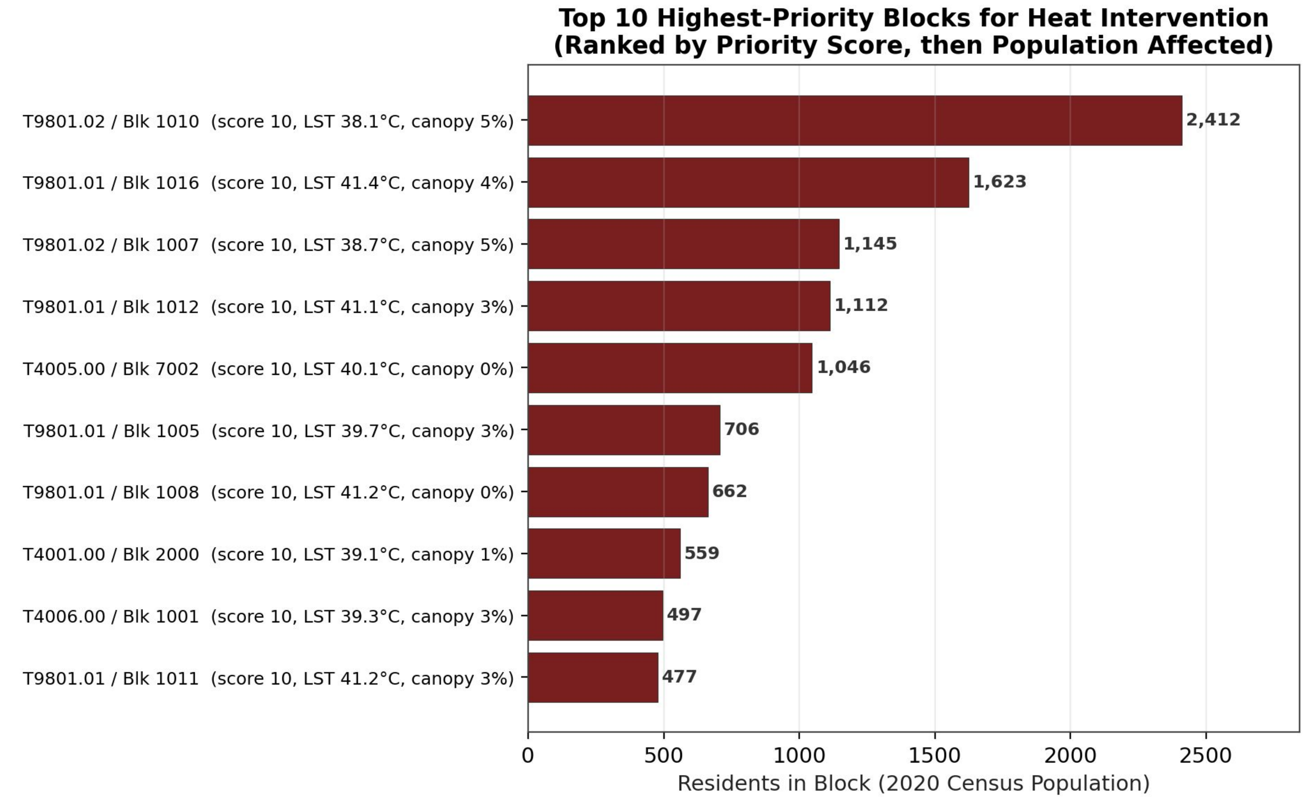

Fig 3. Top 10 highest-ranked census blocks for heat-mitigation intervention, ranked by Priority Planting Score and, among ties, by population affected.

- A Priority Planting Score was developed by combining Land Surface Temperature (LST), impervious surface coverage, tree canopy cover, and street tree density.
- The score identified census blocks with the greatest need for additional tree canopy based on existing environmental conditions.
- High-priority blocks were characterized by high surface temperatures, extensive impervious surfaces, sparse tree canopy, and low street tree density.
- The analysis revealed clusters of priority blocks throughout Ann Arbor, with several concentrated in the downtown core, where opportunities for heat mitigation are greatest.

- Identified the three hottest census blocks in Ann Arbor using satellite-derived Land Surface Temperature (LST).
- All three blocks had more than 88% impervious surface coverage and little to no tree canopy, demonstrating the strong relationship between vegetation loss and elevated surface temperatures.
- The hottest blocks overlap with downtown streets that are pedestrianized during Ann Arbor's Summer Streets program, where roads are temporarily closed to vehicles to encourage walking, dining, and community events.
- These locations experience high pedestrian activity during the warmest months, making them important targets for heat mitigation strategies such as expanded tree canopy, shade structures, and other cooling infrastructure.
- The findings highlight how GIS can identify locations where environmental conditions and human activity intersect.

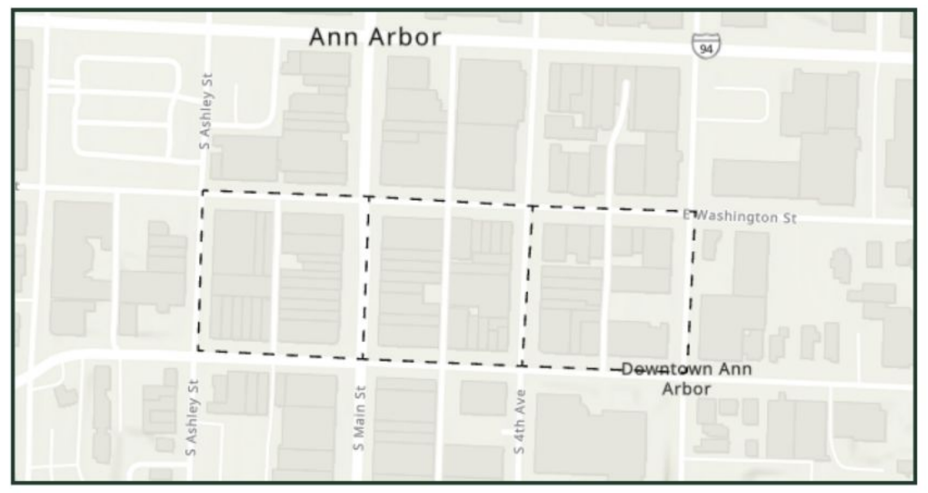

Fig 4. Aerial views of the 3 hottest census blocks

Discussion

- Tree canopy showed the strongest relationship with lower land surface temperatures (LST) across Ann Arbor.
- The Priority Planting Score identified census blocks where additional street trees could provide the greatest cooling benefits.
- Research shows that large-canopy native trees planted over streets and sidewalks can substantially reduce pavement temperatures.
- Reflective ("cool") roofs may benefit densely developed areas with limited planting space but provide fewer long-term environmental benefits than trees.
- Expanding urban tree canopy supports heat mitigation, stormwater management, air quality, biodiversity, and public health.

Conclusion

- GIS and remote sensing successfully identified urban heat patterns across Ann Arbor at multiple geographic scales.
- Higher temperatures were consistently associated with low tree canopy and high impervious surface coverage.
- Three ArcGIS Online dashboards provide interactive tools to support data-driven urban planning and heat mitigation.
- The methodology offers a scalable and reproducible framework that can be updated with future environmental and demographic data.
- Results can help guide urban forestry investments and improve long-term climate resilience.

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